CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Semester – I / II University Examination May 2011

B.Tech. (CL/EC/ME/CE/EE/IT)

CE – 101 Fundamentals of Computing & Programming

Date: 16/05/2011, *Monday* **Time: 1:30 p.m to 4:30 p.m** **Maximum Marks:70**

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1(a) Answer the following questions. **05**

1. What is the significance of main( ) function in a C program?
2. Explain #define.
3. Which are the keywords in switch statement?
4. Explain conditional operator with example.
5. $(320)_{10} = (\quad)_{16}$.

Q.1(b) What is the correct output of the following code? **06**

(1)
```
void main( ){
    int p=6,q ;
    q=p++ * ++p ;
    printf("%d%d",p,q) ;
}
```

(2)
```
void main( ){
    int i;
    for(i=1;i<=5;i++);
      printf("%d",i);
}
```

(3)
```
void main( ){
    int a,b,z;
    a=3+2/3%2;
    b=a++;
    z=sizeof(float) - b;
    printf("%d %d",a,z);
}
```

Q.2 Write programs in C. **12**

1. Write a program to get following output.

```
      4
    4 3
  4 3 2
4 3 2 1
```

2. Write a program to enter and sort the array of 10 integers into ascending order.

OR

Q.2 Write programs in C. **12**

1. Write a program to evaluate the series:
$1/1! + 2/2! + 3/3! + 4/4! + \ldots\ldots\ldots + n/n!$ Use factorial function which receives an integer and returns factorial of an integer argument.
2. Write a program to add 10 integer numbers of an array using pointer variable.(use pointer to enter the elements)

Q-3 Answer the following questions. **12**

1. Explain basic structure of C program with diagram.
2. Differentiate: nesting of *if* vs. *else.... if* ladder statements.
3. Explain C tokens with example.
4. Explain three types of data type with example.

OR

Q-3 Answer the following questions. **12**

1. Give names of the operators used in C language.
2. Explain syntax of *for* loop with example.
3. Explain representation of two dimensional arrays in memory.
4. Explain Actual argument and formal argument in a function.

SECTION – II

Q-4 Do as directed. **11**

1. Define Pointer. 1
2. Differentiate: gets( ) and scanf( ) functions. 1
3. Differentiate: Algorithm and Flow Chart. 1
4. Differentiate: union and structure. 2
5. Differentiate: compiler and interpreter. 2
6. float a[5]. Base address is 2000. What is the address of the element stored at a[4]? 2
7. Explain fopen( ) function with modes. 2

Q-5 Write programs in C. **12**

1. Write a program to compare two strings without using string.h header file.
2. Define a structure that describes a student having members roll no, name, mark1, mark2, mark3.Write a program for 10 students to print the total of marks of each student and the highest of them.

OR

Q-5 Write programs in C. **12**

1. A file abc.txt is opened in write mode. Enter data into that file. Copy the data from abc.txt into another file xyz.txt which is opened in write mode.
2. Read basic salary of a person in float and age in integer. If it is less than Rs.12000 than 10% income tax will be deducted from basic salary and 3% rent allowance will be added to salary. Else 12% income tax and 5% rent. Write a program to calculate net salary of a person having in float with 2 digit accuracy.

Q-6 Answer the following questions. **12**

1. Differentiate: call by value and call by address with example.
2. Explain any four string handling functions in detail.
3. Give different types of categories of user defined functions and explain any two.

OR

Q-6 Answer the following questions. **12**

1. Explain recursion with example.
2. Explain various declaration of structure variable.
3. Explain *fseek*( ) and *ftell*( ) functions related to file.
4. What is the correct output of the following code?

```
void main( )
{
 int a,b,*p1,*p2,x,y,z;
 a = 10 ; b = 20 ;
 p1 = &a ; p2 = &b ;
 x = *p1 * *p2 + 10 ;
 y = 4 *  *p2 / *p1 + 10 ;
 *p2 = *p2 + 3 ;
 *p1 = * p1 * *p2 – 6 ;
 printf(" %d\n ",x);
 printf(" %d\n ",y);
 printf(" %d\n ",*p1) ;
 printf(" %d\n ",*p2) ;
}
```

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